

Round 2: Los Santos Heist Simulation Spec

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This spec turns last year's Lokah trading game (the TTBM repo) into a Los Santos heist game for 12 players. Players can now see the whole city and call anyone, but loot still moves only between neighbouring districts, so they must plan delivery chains. A shrinking completion bonus and one Police Raid add pressure. Six changes are needed, mostly in `server/src/game.ts` and `client/src/components/Dashboard.tsx`.

What changes from last year

	Last year (Lokah)	This year (Los Santos)
Theme	Mythic horror, "countries"	GTA-style city, named districts
Loot	Trishula, Gandiva, Vajra, Brahmastra	Cash, Artwork, Gold, Diamonds
Map	Hidden; players see only neighbour buttons	Full city map visible to every player
Chat	Only with neighbours in Phase 1	With anyone, from the start
Trading in Phase 1	Only with neighbours	Still only with neighbours (unchanged)
Relay bonus	Exists in code but never pays out (bug)	Fixed: every middleman in a delivery chain earns +100
Completion bonus	Flat 1,000	Starts at 1,000 and drops 50 per minute, minimum 200
Mid-game event	None	One Police Raid, triggered by the admin
Elimination	Random "Sacrifice" lottery	Not used in this round; ranking decides who goes through

What stays the same: the 12-node network, login as `TT_BM_1` to `TT_BM_12`, drag-to-trade, the chat, Phase 1 and Phase 2, the admin panel, the timer and the leaderboard.

Game rules (what players experience)

Setup

- 12 players, one per district. Each logs in as `TT_BM_<number>`.
- Each player gets a secret **Heist Order**: 3 items to collect, for example 2 Gold and 1 Diamond.
- Each player starts with loot, mostly items they don't need.
- The game creates exactly as many items as all Heist Orders need (36 for 12 players). Before the raid, everyone can finish if everyone cooperates.

The city map

District	Node	Role	Neighbours
Vinewood	1	Hub	Rockford Hills, Mirror Park, Vespucci
Downtown	2	Hub	Mirror Park, Vespucci, La Mesa
Rockford Hills	3	Bridge	Vinewood, Davis, Strawberry
Mirror Park	4	Bridge	Vinewood, Downtown, Sandy Shores
Vespucci	5	Bridge	Vinewood, Downtown, Paleto Bay
La Mesa	6	Bridge	Downtown, Grapeseed, Chumash
Davis	7	Outlier	Rockford Hills
Strawberry	8	Outlier	Rockford Hills
Sandy Shores	9	Outlier	Mirror Park
Paleto Bay	10	Outlier	Vespucci
Grapeseed	11	Outlier	La Mesa
Chumash	12	Outlier	La Mesa

The connections are the existing ones in `topology.ts`; only the names are new.

Phase 1: Turf War

- Players see the full map and can **chat with anyone**.
- Loot can only be **handed to a direct neighbour**. To get an item from across the city, players arrange a chain of middlemen.
- Every middleman in a chain earns **+100 (Fixer's Cut)** when the item reaches someone who needs it.

Police Raid (once, triggered by the admin)

- The 3 players holding the most loot that isn't locked in a completed heist each lose 1 random item. Seized items leave the game.
- Every screen flashes a raid alert, and the public screen shows a Raid Report listing the raided districts.
- After the raid, up to 3 Heist Orders can no longer be completed. Those players can still score by relaying for others.

Phase 2: Open City

- The admin switches it on for the last few minutes. Anyone can hand loot directly to anyone.

Winning

- Highest score wins. Completing fast and relaying for others both count.

Scoring

A player's score is recalculated by `calculateScore` in `server/src/game.ts` (line 766). The new formula:

```
score = itemPoints + completionBonus + fixerPoints
```

Part	Rule	Change from current code
<code>itemPoints</code>	+100 for each held item that is in the Heist Order, capped at what the order needs (max 300)	None
<code>completionBonus</code>	If the order is complete: $\max(200, 1000 - 50 \times \text{whole minutes from game start to completion})$. Otherwise 0	Was a flat 1,000
<code>fixerPoints</code>	$+100 \times \text{facilitationCount}$	None in the formula; the counting is fixed in Change 3

Completion bonus by minute:

Completed during minute	Bonus
0	1,000

Completed during minute	Bonus
4	800
8	600
12	400
16 or later	200

Maximum possible without relaying: $300 + 1,000 = 1,300$.

Round timeline and admin run sheet

The round lasts 20 minutes of play. Times are from the moment the admin presses Start.

Time	What happens	Admin action
Before	12 players log in; tutorial explains the rules	Check all 12 districts show online on the admin map
0 min	Turf War starts: map visible, chat with anyone, hand loot only to neighbours	Press Start Game , set the timer to 20 minutes and start it
8 min	Police Raid	Press Police Raid (works once)
8 to 16 min	Chains are rebuilt; some heists are now impossible	Nothing; watch the admin map
16 min	Open City : anyone can hand loot to anyone	Press Open City (the existing Initiate Phase 2 button, renamed)
20 min	Final whistle	Press Stop Game ; show the leaderboard on the big screen

The admin should avoid pausing, because the completion bonus counts real minutes (see Change 4).

Change 1: Visible city map

Goal: Every player sees all 12 districts and the roads between them, and knows which districts they can hand loot to.

Today: `Dashboard.tsx` (about lines 336 to 400, the "Lokah Map" section) shows a grid of buttons for neighbours only (`visibleNodes` , line 220). The full network is drawn only in `AdminPanel.tsx`.

What to build

1. Move `INITIAL_POSITIONS` from `AdminPanel.tsx` (lines 17 to 24) into `client/src/topology.ts` as `NODE_POSITIONS`, and import it in both places.
2. Create `client/src/components/CityMap.tsx`:
 - Draw an SVG with `viewBox` sized to the positions (about 0 0 700 580) so it scales on phones and laptops.
 - Draw a road (line) for every neighbour pair, the same way `AdminPanel.tsx` does around line 648.
 - Draw each district as a clickable marker showing the district name, the player alias, an online dot and an unread-message badge.
 - Styles: the player's own district highlighted (yellow); districts they can hand loot to marked "deliverable" (bright border); all others dimmer but still clickable.
 - In Open City (`gameState.stage === 2`) every district is deliverable.
3. In `Dashboard.tsx`, replace the neighbour-button grid with `<CityMap>`. Clicking a district opens the chat with that player, as the buttons do now. Dropping loot on a district calls the existing `handleDrop`.

Keep hidden: other players' loot and Heist Orders. Players must still ask.

Done when: a player in Paleto Bay sees all 12 districts and the roads, with Vespucci marked deliverable in Turf War, and every district marked deliverable in Open City.

Change 2: Call anyone, deliver through your turf

Goal: In Turf War, a player can chat with any district but can hand loot only to a direct neighbour.

Today: The server's `chat_message` handler (`server/src/index.ts`, line 199) already accepts messages to anyone. Only the client limits chat, through `visibleNodes` in `Dashboard.tsx`. The server already blocks non-neighbour transfers in Phase 1 (`transferResource`, `game.ts` line 257).

What to build

1. Chat: with Change 1 in place, any district can be selected, so chat works with anyone. No server change needed.
2. Block loot to non-neighbours in the UI during Turf War:
 - In the chat panel, disable the attach (paperclip) button when the selected district is not a neighbour and `stage === 1`.
 - In `handleDrop` (`Dashboard.tsx`, line 197), refuse drops on non-neighbours in stage 1.

- In both cases show: "Out of your turf. Hand it to a neighbour to pass it on."
- 3. Fix the misleading success toast: `sendMessage` (`Dashboard.tsx` , line 175) shows "ITEM SENT" before the server answers. Show success only when the server accepts. The server already emits `action_error` on failure; add a success acknowledgement or show the toast when the item leaves the player's inventory in the next `state_update` .
- 4. Keep the server check in `transferResource` as the final guard.

Done when: in Turf War, Paleto Bay can chat with Davis but cannot hand Davis loot, and can hand loot to Vespucci. In Open City, Paleto Bay can hand loot to Davis directly.

Change 3: Fixer's Cut (fix the relay bonus)

Goal: Every middleman who passes an item along a chain earns +100 once the item reaches a player who needs it.

Bug in the current code: the relay bonus never pays. In `transferResource` (`game.ts` , line 275) the receiver is added to `token.history` before `getPreviousHolder` runs (line 279). So the "previous holder" it finds is the sender, and `isFacilitation(sender, sender, target)` is always false. A test relay Davis (7) → Rockford Hills (3) → Vinewood (1), where Vinewood needed the item, recorded 0 facilitations.

The old rule would also pay only the last middleman in a chain, and only in Phase 1. Replace it rather than patch it.

New rule

1. Add `transit: string[]` and `paidFixers: string[]` to `ResourceToken` (server and client `types/game.ts`), both starting empty. Items handed out at game start begin with empty lists.
2. In `transferResource` , after the token moves from sender to target:
 - If the item **helps the target's Heist Order** (use the existing `wouldHelpContract`): pay every alias in `token.transit` +100 (`facilitationCount++` , add the token id to `facilitatedTransfers` , push a `Facilitation` record for the admin stats), add them to `token.paidFixers` , then clear `token.transit` .
 - Otherwise: the target is now carrying the item for someone else, so add the **target** to `token.transit` .
3. Anti-abuse guards:
 - Pay nothing if the target has held this token before (their alias appears earlier in `token.history`). This stops two players bouncing an item back and forth.
 - Skip any alias already in `token.paidFixers` , so a player is paid at most once per item.

- Never pay the target itself.
- 4. Recalculate the score of every player who was paid, not only sender and target, then call `syncLeaderboard`.
- 5. Apply the rule in both Turf War and Open City.
- 6. Remove or retire `isFacilitation` and `getPreviousHolder`.

Example: Sandy Shores (9) → Mirror Park (4) → Vinewood (1) → Vespucci (5) → Paleto Bay (10), who needs it. Mirror Park, Vinewood and Vespucci each get +100 when Paleto Bay receives it.

Done when: that chain gives exactly 3 Fixer's Cuts, the admin Stats tab lists them, and bouncing an item back and forth between two players pays nothing.

Change 4: Time Is Money

Goal: The completion bonus shrinks by 50 points per minute, from 1,000 down to a floor of 200, so fast deals beat slow perfect ones.

What to build

1. In `calculateScore` (`game.ts`, line 784), replace `if (complete) score += 1000;` with the time-based bonus from the Scoring section. Use the player's `completionTime` and `state.startTime`.
2. Exclude paused time: add `totalPausedMs` and `pausedAt` to `GameState`; update them in `pauseGame` and `resumeGame`; subtract `totalPausedMs` when working out minutes elapsed. Reset both in `resetGame`.
3. Fix the order in `transferResource`: today the target's score is calculated (line 286) before `completionTime` is set (line 290), so a fresh completion would score with no time. Set `completionTime` first, then calculate scores.
4. Handle a lost completion: if a player's order stops being complete (they hand away a needed item), set their `completionTime` back to `null`. If they complete again later, the new, later time counts. Check both the sender and the target on every transfer.
5. On the player screen, next to the Heist Order, show a live line: "Mission Passed bonus right now: 750", updated every second from `startTime`.
6. In the completion modal (`Dashboard.tsx`, around line 694), show the bonus the player locked in.

Done when: a player who completes at minute 6 scores 700 as their bonus, and the on-screen bonus counts down in steps of 50 each minute.

Change 5: Police Raid

Goal: One admin button that seizes 1 item from each of the 3 players holding the most exposed loot, punishing hoarding and breaking some delivery chains.

Which loot is exposed

- A player whose Heist Order is **not complete**: all their items are exposed.
- A player whose Heist Order is **complete**: only items beyond what the order needs are exposed. A finished heist is "money in the bank" and is never broken by the raid.

Server: new `policeRaid()` method in `GameManager ( game.ts )`

1. Refuse if the game is not `ACTIVE`, is paused, or a raid already happened.
2. For each player, count their exposed items. Rank players by that count, highest first; break ties at random. Skip players with 0 exposed items.
3. Take the top 3. For each, remove 1 random exposed item from their inventory.
4. Log each seizure as a `Transaction` with `to: 'POLICE'`. Seized items leave the game for good.
5. Also remove the seized token id from anyone's pending chains; clearing its `transit` is enough, because the token no longer exists.
6. Recalculate the scores of raided players and call `syncLeaderboard`.
7. Save to state: `raid: { done: true, time: <timestamp>, districts: [<nodeIds>], seized: [{ nodeId, type }] }`. Add `raid` to `GameState` with `done: false` by default, and reset it in `resetGame`.
8. Call `saveState()`.

Server: socket event (`index.ts`)

- Add `socket.on('admin_police_raid', ...)`. Only the admin may trigger it (check the sender's alias is `admin`). Call `policeRaid()`, then `broadcastState()`, then `io.emit('police_raid', { districts })`.

Admin panel (`AdminPanel.tsx`)

- Add a **Police Raid** button next to the Phase 2 button (around line 630), with a confirm dialog. Disable it after it's used and show "Raid done at mm:ss".

Player screen (`Dashboard.tsx`)

- On `police_raid`, show a full-screen red-and-blue flash for about 3 seconds with "LSPD RAID".
- Raided players also get a toast: "LSPD seized your Gold".
- Everyone else gets: "Raid hit Vinewood, Mirror Park and Sandy Shores".

Public screen (`PublicLayout.tsx`)

- Once `raid.done` is true, show a Raid Report banner listing the raided districts, not the items, for the rest of the game.

Done when: pressing the button once removes exactly 3 items from the right players, a completed player's needed items are never taken, and a second press does nothing.

Change 6: GTA reskin

Loot types

Rename the four resource types everywhere. The server and client must match, or trades break.

Old	New	Emoji	Colour
Trishula	Cash		Green <code>#3DDC84</code>
Gandiva	Artwork		Purple <code>#A855F7</code>
Vajra	Gold		Yellow <code>#FFD400</code>
Brahmastra	Diamonds		Cyan <code>#22D3EE</code>

Files: `server/src/types/game.ts`, `server/src/game.ts` (`RESOURCE_TYPES` and every `{ Trishula: 0, ... }` literal), `client/src/types/game.ts`, `Dashboard.tsx` (lines 15 to 32 and 215), `AdminPanel.tsx` (lines 35 to 51 and 477), and the scripts in `server/scripts/`.

District names

- Set the `label` of each node to the district names in the Game rules table, in both `client/src/topology.ts` and `server/src/topology.ts`. Keep `ids` and `neighbours` unchanged.
- Everywhere the UI shows `COUNTRY {id}`, show the district name instead (mainly `Dashboard.tsx`, about 10 places, and `AdminPanel.tsx`).
- In `syncLeaderboard` (`game.ts`, line 344), set `country` to the district name instead of `Node {id}`.

Text

Old	New	Main file
SCARCITY	LOS SANTOS	Lobby.tsx
PROTOCOL INITIATED. / RESOURCE SCARCITY: CRITICAL. / TRUST IS YOUR ONLY CURRENCY.	THE CITY IS OPEN. / PLAN THE JOB. MOVE THE LOOT. / EVERY MINUTE COSTS YOU.	Lobby.tsx
IDENTITY VERIFICATION / ENTER LOKAH	CREW ID / START THE JOB	Lobby.tsx
SYS.VER.2.0.4 // MYTH_CORE	GRAND MANAGER // LOS SANTOS	Lobby.tsx , Leaderboard.tsx
LOKAH DOMINION	LOS SANTOS	Hub.tsx , PublicLayout.tsx
Sacred Contract / AWAITING PROPHECY...	Heist Order / AWAITING THE BRIEFING...	Dashboard.tsx
Ascension Ready	Ready: Mission Passed	Dashboard.tsx
Artifacts / Select Artifact / Attaching Artifact	Loot / Select Loot / Attaching Loot	Dashboard.tsx
VOID / ARTIFACT STORAGE EMPTY	EMPTY / NO LOOT	Dashboard.tsx
Lokah Map	Los Santos	Dashboard.tsx
SECURE_LINK // COUNTRY_x	PHONE // <district>	Dashboard.tsx
SACRIFICE <type> ... to COUNTRY x?	Hand <type> to <district>?	Dashboard.tsx
DISCONNECT FROM LOKAH?	LEAVE THE JOB?	Dashboard.tsx
PHASE 1 STARTED toast	TURF WAR: call anyone, hand loot only to neighbours	Dashboard.tsx
PHASE 2 STARTED toast	OPEN CITY: hand loot to anyone	Dashboard.tsx
Header "Phase 1 / 2"	"Turf War" / "Open City"	Dashboard.tsx
INITIATE PHASE 2 (admin)	OPEN CITY	AdminPanel.tsx

Old	New	Main file
Zero-Sum Ascension Achieved	MISSION PASSED with the bonus earned	Dashboard.tsx
LOKAH TIMER // VER.1.0.0	HEIST CLOCK	Timer.tsx
INITIALIZING MYTH_CORE...	LOADING LOS SANTOS...	Dashboard.tsx
<title>Dominion</title>	<title>Los Santos</title>	client/index.html

Tutorial

Rewrite the 8 steps in `Tutorial.tsx` to match the new rules: Heist Order, loot, the city map, call anyone, hand loot only to neighbours, Fixer's Cut, the shrinking bonus, and that a Police Raid will happen.

Look

- **Colours** (`tailwind.config.js`): replace the horror red accent with GTA yellow `#FFD400` and money green `#3DDC84` ; keep black and white.
- **Fonts**: replace the Cinzel serif with a heavy display font such as Bungee or Bowlby One (Google Fonts). Don't use Pricedown, the GTA font, as it needs a licence.
- **Background** (`.lokah-bg` in `index.css`): replace the moon and stars with a dusk gradient (dark blue to orange) or a city skyline.
- **Headers**: replace the glitch animation with a bold black drop shadow.
- **Map image**: `client/public/network_diagram.png` has old labels and isn't used anywhere. Delete it.

Not used this round

The Sacrifice lottery is not part of this round. Hide its link on the public screen or leave it unused. Don't delete it.

Setup notes and things to leave alone

Working setup

- Work on a new branch, for example `gta-theme` , so last year's version stays on the main branch.
- Run `npm install` and then `npm run dev` from the repo root. The client runs on port 5173 and the server on 3000.

- If Vite fails with "no such file ... react/index.js", delete `client/node_modules/.vite` and restart.
- `server/.env` points `DATABASE_URL` at the live Railway Postgres database. For local testing, run with a dummy `DATABASE_URL` so tests don't write to it. The game still runs; the server just logs database errors. Never commit `.env`.
- `server/gameState.json` is committed to git and holds the saved game. After renaming the loot types, reset it (admin **Reset Game**, or delete the file) so old item names don't load.

Keep as is

- The 12 nodes and their neighbours in `topology.ts`. Only the labels change.
- The login format `TT_BM_<number>` (checked in `game.ts`, line 141).
- Game config: `totalResources: null` and `resourcesPerPlayer: 3`, so supply matches demand exactly. Don't lower supply; the raid creates the scarcity.
- Phase 1 and Phase 2 logic, the timer, pause and stop controls, and the leaderboard.

Test checklist

Run these with 12 browser tabs, or with `server/scripts/bot_swarm.ts` (set `NUM_BOTS` to 11 and leave one district free) plus one real player tab. Update the bot script for the new loot names first.

- ☐ All 12 players can log in, and each sees the full city map with their own district highlighted
- ☐ In Turf War, a player can chat with a non-neighbour
- ☐ In Turf War, handing loot to a non-neighbour is blocked in the UI and by the server, with the "Out of your turf" message
- ☐ In Turf War, handing loot to a neighbour works
- ☐ Relay Sandy Shores → Mirror Park → Vinewood → Vespucci → Paleto Bay (who needs it) pays exactly 3 Fixer's Cuts
- ☐ Bouncing an item between two players pays no Fixer's Cut
- ☐ Completing at minute 6 gives a 700 bonus; the on-screen bonus counts down by 50 each minute
- ☐ Handing away a needed item after completing removes the bonus; completing again uses the new time
- ☐ Police Raid removes exactly 3 items, never touches a completed player's needed items, and works only once
- ☐ Raid alert shows on every player screen; raided players see what they lost; the public screen shows the Raid Report

- ☐ Open City lets anyone hand loot to anyone
- ☐ Leaderboard shows district names and updates after trades and after the raid
- ☐ Reset Game clears scores, completion times, raid state and the paused-time counter
- ☐ Refreshing a player's browser mid-game reconnects them to the same district with their loot
- ☐ Map and chat work on a phone-sized screen
- ☐ No old words left: search the client for Lokah, Country, Artifact, Sacred, Trishula, Gandiva, Vajra, Brahmastra

Open questions

- ☐ Should the raid take 3 items, or scale with the number of players still active?
- ☐ Is 20 minutes the right length, or should the raid and Open City times move?
- ☐ Where does the public screen run on the day (projector link to `/leaderboard` or `/timer`)?
- ☐ Who plays admin on the day, and do they need a practice run?
- ☐ Deadline for the finished build, leaving a day for a full 12-player test